

Proposed Solution Template

Date	29 September 2026
Team ID	SWTID-2026-1542
Project Name	AnswerBot – AI-powered FAQ and Customer Support Automation
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Customer support teams handle a huge backlog of repetitive queries, yet FAQ documentation is created, categorised and updated manually, which is slow and error-prone. Support managers (e.g., Priya) also lack visibility into search trends and quick query resolution, and there is no secure permission layer to stop unauthorised edits to documentation.
2	Idea / Solution description	AnswerBot is a secure REST API platform built with Node.js, Express.js and MongoDB (Mongoose) that automates FAQ creation and support answers. Google Gemini AI generates questions, answers, categories and topic suggestions. Users can search FAQs by keyword/semantic match, and creators get full CRUD control over their own FAQs. JWT handles authentication and route protection; bcrypt encrypts passwords; centralized error handling and schema validation keep the system reliable. Role-based access covers Admin, Content Creator, Authenticated User and Public User.
3	Novelty / Uniqueness	Combines AI-generated FAQ content (Google Gemini) with semantic search and secure role-based authorship control in a single backend. MVC-based, headless API design decouples business logic from any frontend, so it can plug into websites, chatbots or mobile apps. Only original authors (or admins) can modify FAQs, protecting private documentation from public edits.
4	Social Impact / Customer Satisfaction	Customers get instant, accurate answers through searchable FAQs, reducing waiting time and repeated tickets.

S.No.	Parameter	Description
		Support staff save time on repetitive drafting and can focus on complex issues, improving service quality. Makes reliable self-service information accessible to everyone, and is useful for small businesses, colleges and NGOs with limited support staff.
5	Business Model (Revenue Model)	Subscription (SaaS): free tier with limited AI generations; paid tiers for higher usage, more creators and analytics. API usage-based pricing for businesses integrating the FAQ/AI service into their own apps. Enterprise plans: custom deployment, admin dashboards, priority support and integrations (helpdesk, chat widgets).
6	Scalability of the Solution	Stateless JWT sessions and a modular Express architecture allow horizontal scaling behind a load balancer. MongoDB indexing and sharding/replication support growth in FAQ volume and users in real time. AI layer is a separate service integration, so it can scale or switch models independently; the API can later be extended with multi-language support, analytics and new channels.